

Solution Requirements - HouseRent MERN

Date: 18 June 2026

Project Name: House Rent

Maximum Marks: 4 Marks

SOLUTION REQUIREMENTS SPECIFICATION

FUNCTIONAL REQUIREMENTS:

FR-1 User Authentication

- Tenant registration with email, password, name, location
- Owner registration with email, password, name, location
- JWT token generation on successful login
- Email validation and password strength requirements
- Session management and logout

Priority: CRITICAL

FR-2 Property Search & Filtering

- Search by location (city/area name)
- Filter by rent amount (min and max)
- Filter by property type (house, apartment, room)
- Filter by furnishing status (furnished, semi, unfurnished)
- Filter by amenities (parking, pool, gym, WiFi)
- Sort results by price, date, rating

Priority: CRITICAL

FR-3 Property Details

- Display property title, description, location
- Show rent, property type, furnishing status
- List amenities with descriptions
- Display property images
- Show owner name and contact info
- Show availability status

Priority: CRITICAL

FR-4 Owner Property Management

- Add new property listing with all details
- Edit existing property information
- Delete property listings
- Set property availability status
- Upload and manage property images

Priority: CRITICAL

FR-5 Booking System

- Renter submit booking with start/end dates
- Booking gets initial "pending" status
- Owner view and approve/reject bookings
- Status workflow: pending → confirmed → completed
- Renter view real-time status updates
- Property availability updates based on booking

Priority: CRITICAL

FR-6 Admin Approval Workflow

- Admin view pending owner registrations
- Admin approve owner accounts
- Approved owners can list properties
- Unapproved owners blocked from listing
- Owner notified of approval status

Priority: CRITICAL

FR-7 Role-Based Dashboards

- Renter Dashboard: bookings, history, preferences
- Owner Dashboard: properties, inquiries, bookings

- Admin Dashboard: users, properties, statistics
- Priority: HIGH

FR-8 Role-Based Access Control

- Tenants restricted to browsing and booking only
- Owners can only manage their own properties
- Admins have full platform access
- Unauthorized access blocked with 403 errors
- UI reflects role-based permissions

Priority: CRITICAL

FR-9 Data Persistence

- All user data saved to MongoDB
- Property listings persistent
- Booking history maintained
- Approval status saved permanently

Priority: CRITICAL

NON-FUNCTIONAL REQUIREMENTS:

NFR-1 Usability

- Intuitive navigation with clear menus
- Consistent UI design across pages
- Form validation with helpful errors
- Responsive design for desktop/tablet

Priority: HIGH

NFR-2 Security

- Passwords hashed with bcrypt (never plain text)
- HTTPS in production
- JWT tokens with expiration
- CORS configured
- Input validation prevents injection attacks

Priority: CRITICAL

NFR-3 Reliability

- 99% uptime target
- Database connection pooling
- Error handling and logging
- Graceful error recovery

Priority: HIGH

NFR-4 Performance

- Page loads in <3 seconds
- API responses within 200-500ms
- Search results update within 1 second
- Database queries optimized with indexes

Priority: HIGH

NFR-5 Scalability

- Horizontal scaling with load balancing
- MongoDB horizontal scaling support
- Stateless backend design
- Support 1000+ concurrent users

Priority: MEDIUM

NFR-6 Maintainability

- Clean, organized code structure
- MVC pattern implementation
- API documentation
- Version control with Git

Priority: MEDIUM